

Moein Razavi

Senior Software Engineer | Python, Distributed Data Systems, Machine Learning & Full-Stack

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Professional Summary

Software engineer with a Ph.D. and 8+ years of experience across industry and applied research, including 3+ years at Ford designing, testing, launching, and maintaining production software on Google Cloud Platform. Writes object-oriented Python across backend services, distributed data pipelines (10+ TB), and machine learning systems, with full-stack and mobile delivery in TypeScript/React and Swift. Leads design decisions on projects from ambiguous requirements to launch, builds testing and validation into production systems, and has launched products as a founder. Experience spans information retrieval, natural language processing, distributed computing, security, and mobile.

Professional Experience

Ford Motor Company

Senior Data Scientist | Aug 2023 - Present

- **Distributed data pipelines:** Built and maintain EV telemetry pipelines processing 10+ TB across 15 Airflow DAGs using PySpark on Dataproc, BigQuery, and Dataform; debug production data issues end to end, including time-of-use rate and savings calculations used in smart-charging reports.
- **Multi-agent retrieval and question-answering system:** Reduced time spent answering recurring analytics and document questions by 30% by designing a multi-agent service with intent routing, BM25 retrieval-augmented generation over legal documents and vehicle manuals, and schema-aware text-to-SQL over BigQuery; added domain-rule answer checks to catch incorrect output and launched on Cloud Run behind SAML/ADFS authentication.
- **Automated ingestion service (HOMeload):** Reduced manual data preparation time by approximately 50% by designing a 5-agent Python/FastAPI service for inconsistent third-party utility bills, with PII redaction, deterministic validation, bounded retries, and idempotent BigQuery MERGE loads; deployed with Docker, Cloud Run, and Terraform.
- **Code review tooling:** Reduced manual code-review time by 60% by building a FastAPI service that parses repositories with GitPython and Python AST analysis and streams LLM-assisted complexity reviews over Server-Sent Events, with Vertex AI cost estimates.
- **Vehicle-to-grid market bidding:** Reduced vehicle-availability forecast error by approximately 14% versus the previous baseline across approximately 9,500 vehicles by implementing custom PatchTST-style transformers in PyTorch, benchmarking TimesFM, MOIRAI-2, and iTransformer, and tuning with Optuna on Vertex AI GPUs; the capacity estimates, built with Monte Carlo simulation and greedy vehicle selection, support ERCOT market bids.
- **Backend services and CI/CD:** Designed Flask REST APIs backed by BigQuery for XGBoost bill predictions and a SAML-authenticated Dash front end, launched on Cloud Run through Tekton CI/CD.

CereVu Medical, Inc.

Software Developer / Research Engineer (Concurrent Contract) | Jun 2023 - Sep 2024

- **Mobile and full-stack delivery:** Developed and launched two SwiftUI iOS apps (Vitality and Clinical) capturing 9 physiological signals over CoreBluetooth BLE, distributed through TestFlight and App Store submissions, with a Python/Flask, WebSocket, and MongoDB backend feeding a React/Next.js dashboard.
- **Architecture and infrastructure:** Redesigned the Clinical app from UIKit to an MVVM-Coordinator architecture supporting calibration, exports, role-based workflows, and OAuth 2.0/JWT authentication; deployed Docker services on AWS EC2 with GitHub Actions CI/CD.

Education

Ph.D., Industrial Engineering | Texas A&M University | Dec 2023 | GPA: 4.0/4.0

Master of Engineering, Computer Engineering | Texas A&M University | May 2021 | GPA: 4.0/4.0

Selected Projects

- **Inspealytics (founder and technical lead):** Designed and launched a product that reduced inspection-report preparation time by approximately 50%, spanning a SwiftUI iOS app, a React/TypeScript web app, and Supabase (PostgreSQL); implemented offline edit conflict resolution, ARKit/Apple Vision photo organization, and XCTest/Vitest test suites.
- **Clink Social (founder):** Built SwiftUI/MapKit discovery and messaging on Firebase with TypeScript Cloud Functions for

feeds and notifications, plus QR-based sign-in using time-limited OAuth/JWT tokens.

- **mHELP transformer integration (2026):** Deployed a PyTorch-exported transformer to Apple Watch via Core ML, reimplementing training-time preprocessing in Swift for on-device inference.

Research Experience

Texas A&M University

Research Assistant | Sep 2018 - May 2024

- Published research in machine learning and sensing; 500+ Google Scholar citations (September 2026).
- **OpenSync (open source):** Co-developed and tested open-source Python software (Lab Streaming Layer) that synchronizes multiple sensor streams and experiment events; published tests measured RMS recording-time lag below 0.07 ms.
- **Model evaluation and explainability:** Evaluated heart-rate/motion classifiers on data from 54 participants over 40 days using cross-validation, ROC-AUC, and SHAP to explain model behavior; trained Keras MLPs and contributed to the earlier mHELP scikit-learn and XGBoost inference pipeline.
- **MERCURY (IEEE HPCA 2023):** Co-authored research using random projection with quantization and cached computation reuse; reported 1.97x average DNN training speedup across 12 models at baseline accuracy.
- **Computer vision:** Developed face-mask and physical-distance monitoring on video using TensorFlow Faster R-CNN detectors and calibrated perspective transforms.

Technical Skills

Languages: Python, SQL, TypeScript/JavaScript, Swift, Java, C++

Software Engineering: Object-oriented design, data structures and algorithms, API design, testing (XCTest, Vitest), code review, CI/CD (Tekton, GitHub Actions), debugging production systems

Backend & Distributed Systems: FastAPI, Flask, REST, WebSocket, Server-Sent Events, PySpark, Dataproc, Airflow, BigQuery, PostgreSQL, MongoDB, Firebase

Cloud & Infrastructure: Google Cloud Platform (Cloud Run, Vertex AI, Dataproc, BigQuery, Dataform), AWS (EC2, S3, Lambda), Docker, Kubernetes, Terraform

Machine Learning & AI: PyTorch, TensorFlow/Keras, scikit-learn, XGBoost; transformers, information retrieval (RAG, BM25), NLP, LangChain, LangGraph, Google ADK

Frontend & Mobile: React/Next.js, SwiftUI, UIKit, Core ML, CoreBluetooth, ARKit

Security: OAuth 2.0, JWT, SAML/ADFS, role-based access control, PII redaction

Selected Publications

Razavi, M., et al. Machine Learning, Deep Learning, and Data Preprocessing Techniques for Detecting, Predicting, and Monitoring Stress and Stress-Related Mental Disorders: Scoping Review. *JMIR Mental Health*, 2024.

Janfaza, V., Weston, K., Razavi, M., et al. MERCURY: Accelerating DNN Training By Exploiting Input Similarity. *IEEE HPCA*, 2023.

Razavi, M., et al. OpenSync: An open-source platform for synchronizing multiple measures in neuroscience experiments. *Journal of Neuroscience Methods*, 2022.